IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc

GAME THEORY

Time allowed: 3:00 hours

There are FIVE questions on this paper.
Answer FOUR questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 2.

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Second Marker: D. Angeli

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Information for candidates:

No special instructions for candidates.

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1. Take an $n \times m$ matrix S. Consider a zero sum matrix game in mixed strategies:

$$\underset{y \in \Lambda_n}{\text{Min}} \ \underset{z \in \Lambda_m}{\text{Max}} \ y^T S z \tag{1}$$

in which, for a given integer r, Σ_r denotes the set of probability mass functions

$$\Lambda_r := \{(p_1, \ldots p_r) \, : \, \sum_i p_i = 1 \text{ and } p_i \geq 0 \text{ for } i = 1, \ldots, r\} \, .$$

(a): Assume that S is the matrix

$$\begin{bmatrix} 6 & 5 & 7 & 0 \\ 5 & 7 & 0 & 8 \end{bmatrix} .$$

By graphical means, or otherwise, derive a Nash equilibrium $(\bar{y}, \bar{z})$ for the game. [12]

Evaluate the pay-off V. [2]

(b): Let $(\bar{y}, \bar{z})$ be a Nash equilibrium for (1), with pay-off V. Show that $(\bar{y}, \bar{z}, \bar{p} := -V, \bar{q} := V)$ is a solution to the optimization problem

$$(P) \begin{cases} \text{Minimize } p + q \\ \text{over scalars } p \text{ and } q \text{ and vectors } y \in \Lambda_n \text{ and } z \in \Lambda_m, \\ \text{subject to} \\ Sz \geq -p 1_n \text{ and } S^T y \leq q 1_m \, , \end{cases}$$

in which, for a given integer r, 1_r denotes the r dimensional vector $[1, \ldots, 1]^T$. [6]

Annotations: Part (b) is bookwork from the course notes. As shown in the course notes, there is precise equivalence between the Nash Equilibria in mixed strategies and solutions to the linear program (P). Some students wasted time trying to show this equivalence, when they were asked only to show the (one-way) implication '$(\bar{y}, \bar{z})$ be a Nash equilibrium for (1), with pay-off V' implies '$(\bar{y}, \bar{z}, \bar{p} := -V, \bar{q} := V)$ is a solution (P)'.

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2. Consider the 2 stage zero sum game summarized by the two diagrams Fig. 1(a) and Fig. 1(b), involving players A and B, and also a third player N ('Nature'), who plays a fixed random strategy known to players A and B. Players A and B aim to minimize and maximize the pay-off, respectively.

A acts first and B acts second. There is a communication link between the players, providing information about A's strategy to B. Nature determines, with equal probability, whether, or not, there is a fault on the link, which prevents the information being transmitted. Thus,

Nature chooses Scenario 1 ('no fault') w.p. 1/2 and Scenario 2 ('fault') w.p. 1/2, where

Scenario 1 (no fault): A knows B's action as in Fig. 2(a)

Scenario 2 (fault): A does not know B's action, as in Fig. 2(b)

Reduce the game to normal form, according to the following method.

Step 1: Identify the 8 possible strategies, $(B_1)\ldots,(B_8)$ available to B, depending on the actions of A and Nature. [10]

Hint: A representative strategy for B, (B_1) *say, is*

$$(B_1): B \text{ chooses } \begin{cases} L & \text{if } A \text{ chooses } L, \text{ and Nature chooses 'no fault'} \\ R & \text{if } A \text{ chooses } R, \text{ and Nature chooses 'no fault'} \\ L & \text{if Nature chooses 'fault'} \end{cases}$$

Step 2: Calculate the two 2×8 pay-off matrices, relating to the scenarios 'no fault' and 'fault' and average them according to the probabilities of occurence. [6]

Identify any Nash equilibria, and state whether they are admissible. [4]

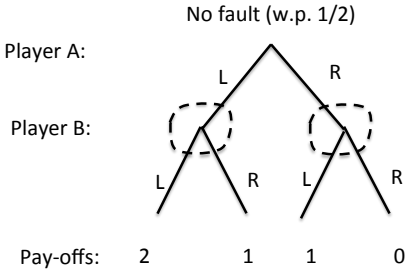


Fig. 2(a)

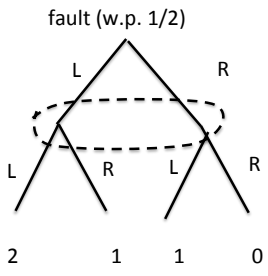


Fig. 2(b)

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3: Consider a game with two players, A and B. Both players aim to minimize their pay-offs. The pay-offs of players A and B are

$$L^A(a,b) := -ab + \tfrac{1}{2}a^2$$

$$L^B(a,b) := \begin{cases} -a^2b + \frac{1}{2}b^2 & \text{if } a \leq 2 \\ -4b + \frac{1}{2}b^2 & \text{if } a > 2\,, \end{cases}$$

respectively. The policy spaces are $\mathcal{A} = \{a : -\infty < a < \infty\}$ and $\mathcal{B} = \{b : -\infty < b < \infty\}$.

Sketch the response curves for players A and B. [4]

Determine all the Nash equilibria, and the pay-offs associated with each equilibrium. [6]

Which is the 'best' equilibrium and in what sense? [2]

Now consider a Stackelberg game for the two players, with the same pay-offs. Assume that A is the leader.

Find a function $\psi(a)$ such that, if $\bar{a}$ is a Stackelberg minimizer, then

$$\psi(\bar{a}) = \min_{a \in \mathcal{A}} \psi(a)\,.$$

[4]

Hence determine any Stackelberg minimizers $\bar{a}$ for A, and the optimal strategies for the B player when A chooses $\bar{a}$. [4]

Annotations: To identify the minimizers of $\psi(.)$ is helpful to sketch the graph. Setting the gradient to zero reveals two local minimizers. The graph shows that only one of these is a global minimizer.

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4. 3 isolated people, X_1, X_2, X_3, live on an island, when a volcano erupts. One person needs to sail to the mainland, to get help.

Because different people have different distances to sail, the pay-off for X_i (which X_i aims to maximize) depends on i. It is:

Satisfaction Level	Scenario
1	Someone (but not X_i) sails to the mainland
$1 - c_i$	X_i sails to the mainland
-1	no one sails to the mainland

(X_i gets high satisfaction (pay-off = 1) if someone sails, but her satisfaction is reduced by an inconvenience factor c_i, if she sails herself. She gets low satisfaction (pay-off = -1) is no one sails.)

Here c_1, c_2, c_3 (the inconvenience factors of X_1, X_2, X_3 respectively) are numbers satisfying

$$0 < c_1 < 1,\ 0 < c_2 < 1,\ 0 < c_3 < 1\,.$$

For $i = 1, 2, 3$, X_i must choose a mixed strategy independent of the other players' strategies.

Suppose that, for each i, X_i chooses a mixed strategy

$$\text{'not sail' w.p. } (1 - \lambda_i),\ \text{'sail' w.p. } \lambda_i$$

while, for $j \neq i$, X_j chooses

$$\text{not sail' w.p. } (1 - \bar{\lambda}_j),\ \text{'sail' w.p. } \bar{\lambda}_j.$$

Derive the expected pay-offs for X_i, $i = 1, 2, 3$. [10]

Hence, or otherwise, derive three equations for $\bar{\lambda}_1, \bar{\lambda}_2$ and $\bar{\lambda}_3$ such that

$$\Big(((1 - \bar{\lambda}_1), \lambda_1), ((1 - \bar{\lambda}_2), \lambda_2), ((1 - \bar{\lambda}_3), \lambda_3)\Big)$$

is a Nash equilibrium in 'strict' mixed strategies, assumed to exist, for the three players. [4]

Now assume that

$$c_1 = c_2 = c_3 = c\,.$$

In the case $0 < c < 2$ solve the equations for the Nash equilibrium in strict mixed strategies. What is the probability that no one will sail? [4]

Is there is Nash equilibrium if $c \geq 2$?

Annotations: Some students manages to obtain explicit formulae for the solutions $(\bar{\lambda}_1, \lambda_2, \lambda_3)$*, though this was not requested.* [2]

Annotations: This question is straightforward, regarding technique, but required a lengthy calculation.

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5. Consider the dynamic zero sum game

$$\begin{cases} \underset{\{u_t\}}{\text{Minimize}}\ \underset{\{v_t\}}{\text{Maximize}}\ \sum_{t=0}^{N-1} L(x_t, u_t, v_t) + g(x_N) \\ \text{over state feedback strategies } \{u_t\}_0^{N-1}, \{v_t\}_{t=0}^{N} \text{ and states} \{x_t\}_{t=0}^{N} \text{ such that} \\ x_{t+1} = f(x_t, u_t, v_t) \ \text{ for } t = 0, \ldots, N-1 \\ x_0 \text{ a given vector}\,. \end{cases}$$

(Here, $f(x,u,v)$, $L(x,u,v)$ and $g(x)$ are given functions of the vector variables (u,v,x).)

(A): Briefly describe the Dynamic Programming technique to determine a saddlepoint, in state feedback form, for this game. [5]

(B): Now assume that x, v and u are scalar variables and

$$f(x,u,v) = \Big(x^3 + u + v\Big)^{1/3},\ L(x,u,v) = u^2 - \gamma v^2 \text{ and } g(x) = x^6\,,$$

in which γ is a given positive number.

What is the Bellman equation for the value function $V_t(x)$, $t = 0, \ldots, N$? [3]

Assume that the value function is of the form

$$V_t = p_t x^6 \text{ for } t = 0, \ldots, N.$$

Assume also that the the zero-sum games appearing on the right side of the Bellman equation have a saddlepoint.

Derive recursive equations for the numbers $\{p_0, \ldots, p_N\}$ in these relations. [8]

Hence show that the Nash equilibrium strategies are of the form

$$u_t = k_t^{(u)} |x_t|^3 \text{ and } v_t = k_t^{(v)} |x_t|^3$$

and express the gains $k_t^{(u)}$ and $k_t^{(v)}$ in terms of the p_t's. [4]

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